

CLASS 9th
PHYSICS PRACTISE SHEET

A. Motion

1. A car starts from rest and accelerates at 2 m/s^2 for 10 s. Find final velocity and distance.
2. A train moves 500 m in 20 s. Find its speed.

B. Force & Laws of Motion

1. A body of mass 5 kg is acted upon by force of 20 N. Find acceleration.
2. Two ice skaters push each other. Skater A = 40 kg, Skater B = 60 kg. Find ratio of accelerations.

C. Gravitation

1. Find force between 2 masses of 10 kg each, 2 m apart. $G = 6.67 \times 10^{-11}$
2. Calculate weight of 50 kg person on Earth and Moon ($g = 9.8, 1.62 \text{ m/s}^2$)

D. Work, Energy & Power

1. A body of 2 kg moves at 5 m/s. Find KE.
2. Lift a 20 kg box to height 5 m. Find work done.

E. Sound & Light

1. Speed of sound in air = 340 m/s. Find wavelength of 170 Hz sound.
2. Find focal length of lens forming image at 20 cm for object at 30 cm.